Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
States value for k.	AO1.1b	B1	$k = \dfrac{1}{b+a}$
Uses correct integral for E(R)	AO1.1a	M1	$E(R) = \int_{-a}^{b} kr \, dr = \left[k\dfrac{r^2}{2}\right]_{-a}^{b}$
Uses limits correctly with 'their' integral	AO1.1a	M1	$= \dfrac{k}{2}(b^2 - a^2)$ $= \dfrac{1}{2(b+a)}(b-a)(b+a)$
Completes clear correct rigorous workings to show required result **AG** Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	$= \dfrac{(b-a)}{2}$ **AG**
			Total 4 marks

Q2.

(a) (i) Just catches a tram

Must refer to the 0 in some way to score the E1

E1

= 2 (+ 0) + 20 + 5 = 27

but can score B1 for

2 + 20 + 5 = 27

B1

(ii) b = 37

B1

3

(b) E(T) = 32

B1

Var(T) = 10^2 / 12

= 100 / 12 = 25 / 3 = 8⅓ = 8.33

Any form

B1

2

(c) (35 − 27) = 8

Or by integration from 27 to 35

M1

× 0.1 = 0.8

A1

2

[7]

Q3.

(a) 21.05 and 21.15

both (allow $21.04\dot{9}$ *and* $21.14\dot{9}$ *)*

B1

1

(b) E(X) = 0 (symmetry)

B1

Var (X) $\frac{1}{12}(0.05 - -0.05)^2 = \frac{1}{12} \times \frac{1}{100}$

M1

For R [−a,a] : E(X) = 0 if a = 0.05, 0.1, 0.5

then: Var(X) = $\frac{1}{12}(a - -a)^2$ **or**

their a = 0.049 to 0.05 used for M1

$\Rightarrow sd(X) = \sqrt{\frac{1}{12} \times \frac{1}{100}} = \frac{1}{20\sqrt{3}}$

or $\frac{\sqrt{3}}{60}$ or $\sqrt{\frac{1}{1200}}$

0.0289 (3sf) A0

A1

3

(c) P(−0.01 ≤ X ≤ 0.03) = 0.04×10

cao from correct value used

= 0.4

$\int_{-0.01}^{0.03} 10\,dx = [10x]_{-0.01}^{0.03}$ *= 0.4* *oe*

B1

1

[5]

Q4.

(a) (i) Area / F(x)=10u × 0.01π (OE)

Shown by any correct method

B1

= 1 $\Rightarrow$ $u = \dfrac{10}{\pi}$

Alternatives: $f = \dfrac{1}{10u}$ *B1*

Bdep1

or

$u = \dfrac{10}{\pi}$ $\Rightarrow$ F(x) = 1

Show $u = \dfrac{10}{\pi}$ ***or*** *show* $\dfrac{1}{10u} = 0.01\pi$ *Bdep1*

(Bdep1)

2

(ii) $\mathrm{E}(X) = \dfrac{1}{2}(11u + u) = 6u = 6 \times \dfrac{10}{\pi} = \dfrac{60}{\pi}$

Must be in terms of π (eg 60π^{-1})

B1

1

$$\mathtt{Var}(X) = \frac{1}{12}(b-a)^2$$

$$\mathtt{Var}(X) = \frac{1}{12}(11u-u)^2$$

$$= \frac{1}{12} \times 100 \times \frac{100}{\pi^2} = \frac{100^2}{12\pi^2}$$

Alternatives:

$$\frac{10000}{12\pi^2} = \frac{5000}{6\pi^2} = \frac{2500}{3\pi^2} = \left(\frac{50}{\pi\sqrt{3}}\right)^2 = \frac{(\mathtt{AWRT}\ 833)}{\pi^2}$$

Must be in terms of π

B1

1

(iii) $C = \pi\left(X + \frac{10}{\pi}\right)$

$$\mathrm{E}(C) = \left.\begin{array}{l} \pi \times [\texttt{their}\ \mathrm{E}(X)] + 10 \\ \pi \times \frac{60}{\pi} + 10 \end{array}\right\}$$

*Their **numerical** value of E(X) used correctly*

Must have a multiplier of π or 2π

M1

= 70

CAO

A1

$$\mathrm{Var}(C) = \pi^2 \times \frac{100^2}{12\pi^2} = \frac{100^2}{12}$$

π^2 × [their Var(X) >0]

Must have a multiplier of π^2 or $4\pi^2$

M1

$= 833\frac{1}{3}(833.\dot{3})$

Alternatives:

$\frac{10000}{12} = \frac{5000}{6} = \frac{2500}{3}$

Must be exact: 833.3 gets A0

A1

4

(b) *n* = 100 and $\bar{a} = 40.5$

95% CI for $\mu = \left.\begin{array}{l} 40.5 \pm z \times \frac{\sqrt{25}}{\sqrt{100}} \\ 40.5 \pm 1.0 \end{array}\right\}$

For z = 1.96

B1

z = 1.96 or 1.64 to 1.65 only

M1

= (39.5, 41.5)

AWRT

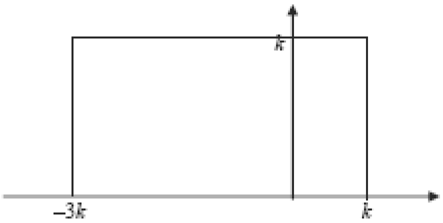
A1

3

[11]

Q5.

(a) (i)



Horizontal line f(x) = k
From –3k to k
If $\frac{1}{2}$ then max. B1

B2,1 **2**

(ii) Area = $4k \times k = 1$

$k^2 = \frac{1}{4}$

SC *If use $k = \frac{1}{2}$ to show that the Area = 1 then $\Rightarrow$ B1*

M1

$k = \frac{1}{2}$ $(k > 0)$

AG

A1 **2**

(b) $\mathrm{E}(X) = \frac{1}{2}(-3k + k)$

$= -k$

$= -\frac{1}{2}$

CAO

B1

$\mathrm{Var}(X) = \frac{1}{12}(k - -3k)^2 = \frac{16k^2}{12} = \frac{4k^2}{3}$

$= \frac{1}{3}$

CAO

M1

st. dev $(X) = \frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$ or $\sqrt{\frac{1}{3}}$

*OE (**exact**)*

A1 **3**

(c) (i) $P\left(X \geq -\frac{1}{4}\right) = \frac{1}{2} \times \frac{3}{4}$

M1

$= \frac{3}{8}$ (0.375)

A1 **2**

(ii) $P\left(X \neq -\frac{1}{4}\right) = 1$

B1 **1**

[10]

Q6.

(a) (i) $k = \frac{1}{a+b}$

B1 **1**

(ii) E(T) $= \int_{-a}^{b} kt\,dt$

M1

$= \left[\frac{kt^2}{2}\right]_{-a}^{b}$

A1

$= \frac{1}{2} \times \frac{1}{(a+b)} \times \left[b^2 - a^2\right]$

$= \frac{1}{2} \times \frac{1}{(a+b)} \times (b-a)(a+b)$

$= \frac{1}{2}(b-a)$

Factorise

M1

AG

A1

4

(b) (i) E(T) = 1

CAO

B1

1

(ii) P(T < –3 or T > 3)

= P(T < –3) + P(T > 3)

M1

= 0.1 + 0.3

= 0.4

A1

Alternative

1 – P (–3 < T < 3)

1 – (0.3 + 0.3) = 0.4

2

[8]

Q7.

(a) For a Rectangular Distribution

$$f(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

(–0.05, 0.05) $\Rightarrow$

(explain error ± 0.05)

B1

$$\frac{1}{b-a} = \frac{1}{0.05-(-0.05)} = \frac{1}{0.1} = 10$$

M1A1

(Area = 10 × 0.1 =1)

3

(b) P(–0.01 < X < 0.02) = 0.03 × 10 = 0.3

M1A1

2

(c) Mean = 0

CAO

B1

Standard deviation = 0.0289

$\frac{1}{20\sqrt{3}}$ *OE*

B1

2

[7]

Q8.

(a) (i) $X \sim \mathrm{R}\,(a, a + k)$

V(X) = $\frac{(a + k + a)^2}{12} = \frac{k^2}{12}$

CAO
use of (b – a) must include = k

B1

= 48 $\Rightarrow$ $k^2 = 576$ $\Rightarrow$ $k = 24$

or $k = 24$ $\Rightarrow$ $\frac{k^2}{12} = \frac{24^2}{12}$ =48

CAO; AG

B1

2

(ii) E (X) = $\frac{a + k + a}{2} = a + \frac{k}{2}$

CAO; OE
use of (a + b) must include = (2a + k)
OE

B1

6 => a + 12 = 6 => a = –6

or a = –6 => a + $\frac{k}{2}$ = *–6 + 12 = 6*

CAO; AG

B1

2

(b) P(X > 0) = $\frac{a + k - 0}{a + k - a} = \frac{a + k}{k}$

attempt at area of a rectangle of height $\frac{1}{k}$

M1

$= \frac{3}{4}$ or 0.75

CAO

A1

2

[6]

Q9.

(a) Mean, $\mu = 21 = \frac{a+b}{2}$

CAO; stated or used

B1

Variance, $\sigma^2 = 27 = \frac{(b-a)^2}{12}$

CAO; stated or used

B1

so
$((42 - a) - a)^2 = 12 \times 27 = 324$
or $b - a = (\pm)\ 18$

Substitution of μ into σ^2 or $\sqrt{\ }$ of equation involving σ^2

M1

Thus $(42 - 2a) = (\pm)\ 18$

Solving quadratic or two simultaneous equations

M1

or $a + b = 42$ and $b - a = (\pm)\ 18$

Thus $a = 30$ or 12 and $b = 12$ or 30

As $a < b$ so $a = 12$ and $b = 30$

CAO; must state a < b

B1 for (12, 30) $\Rightarrow \mu = 21$

B1 for (12, 30) $\Rightarrow \sigma^2 = 27$

A1

5

(b) (i) $P(5 < X < 20) = P(12 < X < 20) =$

Lower limit of 12 or 20 to 30

B1

$\frac{20-l}{b-a}$ or $1-\frac{30-20}{b-a}$

Attempt at area of a rectangle of height $\frac{1}{b-a}$ *or* $\frac{1}{18}$
Can be scored in (ii)

M1

= 8/18 or 4/9 or 0.44
CAO/AWRT; OE

A1
3

(ii) $P\left(X < \mu - \frac{\sigma\sqrt{3}}{2}\right) =$
$P\left(X < 21 - \frac{\sqrt{27}\sqrt{3}}{2}\right) =$
Substitution of $\mu = 21$ *and* $\sigma = \sqrt{27}$*; OE*

M1

P(X < 16.5)
CAO

A1

= 4.5/18 or 1/4 or 0.25
CAO; OE

A1
3

[11]

Q10.

(a) $c = \frac{1}{230-140} = \frac{1}{90}$ or 0.011
cao/awrt

B1
1

(b) P(X < 200) = c × (200 – 140)

M1

= $\frac{2}{3}$ or 0.67
attempt at area of rectangle of height c
cao/awrt

A1 2

(c) Mean: $\mu = \frac{230 + 140}{2} = 185$

cao

B1

Variance $\sigma^2 = \frac{(230 + 140)^2}{12} = 675$

cao

B1 2

[5]

Q11.

(a) (i) Mean = $\mu = 4c$

CAO

B1

Variance = $\sigma^2 = 3c^2$

CAO

B1 2

(ii) $E(X^2) = Var(X) + (E(X))^2$

use of or equivalent

M1

$= 3c^2 + (4c)^2$

A1ft 2

$= 19c^2$

AG

(b) $19c^2 = 171$

$\therefore\ c = 3$

CAO

B1 1

(c) (i) $P\left(X > \frac{\mu}{2} + \frac{\sigma}{\sqrt{3}}\right) = P(X > 6 + 3)$

$= P(X > 9)$

CAO

B1

$= \frac{7c-9}{6c}$ or $1 - \frac{9-c}{6c}$

attempt at correct area

M1

$= \frac{2}{3} = 0.67$

CAO/AWRT

A1

3

(ii) $P(X < d) = 0.25$

$P(X < d) = \frac{d-c}{6c} = \frac{d-3}{18}$

*attempt at correct area **and** substitution of their value of c*

M1

$\therefore \frac{d-3}{18} = 0.25$

Equating their expression in d to 0.25

m1

$\therefore$ d = 7.5

CAO

A1

3

[11]

Q12.

(a) $f(x) = 1/4c \qquad -c < x < 3c$

Mean, $\mu = \frac{-c+3c}{2} = c$

CAO

B1

Variance, $\sigma^2 = \frac{(3c-(-c))^2}{12} = \frac{16c^2}{12} = \frac{4c^2}{3}$

CAO

B1 **2**

(b) $E(X^2) = Var(X) + (E(X))^2 = \sigma^2 + \mu^2$

$\therefore$ *use of, or* $\int x^2$ f(*x*)d*x*

M1

$\therefore \frac{28}{3} = \frac{4c^2}{3} + c^2 = \frac{7c^2}{3}$

substitution of answersin (a) or in intergral plus = $\frac{28}{3}$

m1

$\therefore$ $7c^2 = 28$ or $3c^2 = 12 \Rightarrow c = 2$

CAO; AG

NB accept reverse argument

A1 **3**

(c) (i) P(X = 2) = zero

B1 **1**

(ii) P(|X| < 2.5) = P(–2.5 < X < 2.5)

= P(–2 < X < 2.5)

CAO; seen or implied

B1

$\frac{2.5-(-2)}{8} = 1-\left(\frac{6-2.5}{8}\right)$

attempt at area of a rectangle or attempt at integration

M1

$= \frac{9}{16}$ = 0.562 to 0.563

CAO / AWFW

A1 **3**

[9]

Q13.

(a) E(T) = $\frac{4b-b}{2} = \frac{3b}{2}$

CAO; OE

B1

= 9

Equating their E(T) = 9

M1

2

gives b = 6

AG; discrete $\Rightarrow$ *B0M0*

accept reverse argument, but must state b = 6 gives E(T) = 9

(b) P(T < 15) = $\frac{15-(-6)}{30} = \frac{21}{30}$

worthwhile attempt at correct area and using b = 6

M1

or = 1 $-\left(\frac{24-15}{30}\right) = 1 - \frac{9}{30}$

= $\frac{7}{10}$ or 0.7

CAO; discrete $\Rightarrow$ *M0A0*

A1

2

[4]

Q14.

(a) Mean μ = $\frac{a+(a+k)}{2} = \frac{2a+k}{2}$ or a + $\frac{k}{2}$

B1

Variance σ^2 = $\frac{((a+k)-a)^2}{12} = \frac{k^2}{12}$

CAO

B1

2

(b) Attempt at using results from (a)

M1

$\text{Var}(X) = 3 \Rightarrow \frac{k^2}{12} = 3$ or $k^2 = 36$

or equivalent; must be stated

AG

A1

$\therefore\ k = 6$

$E(X) = 21 \Rightarrow a + 3 = 21$

$\therefore\ a = 18$

or equivalent; must be stated

AG

[Accept reverse arguments]

A1

3

(c) $P(20 < X < 25) = P(20 < X < 24)$

Upper limit of 24 or (a + k)

B1

$= \frac{24 - 20}{k}$

Area of rectangle of height k

M1

$= \frac{2}{3}$ or 0.67

CAO/AWRT

A1

3

[8]

Q15.

(a) $\mu = 201$

CAO (ignore how obtained)

B1

$\sigma^2 = 3$

cao

B1

2

(b) $f(x) = \frac{1}{6}$

CAO; accept AWRT 0.167 (accept h = k = $\frac{1}{6}$ written or on diagram unless followed by f(x) = 6)

B1

1

(c) (i) $P(X = \mu) = 0$

CAO

B1

1

(ii) $P(X < 200 - \sigma) =$

$$\frac{(200-\sigma)-198}{6} = 1 - \frac{(204-(200-\sigma))}{6}$$

use of area of rectangle or integral (allow for 200 – σ or 200 – σ^2 < 198 providing also ∴ prob = 0)

M1

$$= \frac{2-\sqrt{3}}{6} = 1 - \frac{(4+\sqrt{3})}{6}$$

must use their $\sqrt{\sigma^2}$ and their f(x) in formula or integral

A1f.t.

$(2 - \sqrt{3}) / 6 = 0.045$

CAO OE / AWRT

A1

3

[7]

Q16.

(a) So that area = (512 – 500) × $\frac{1}{12}$ = 1

Area = 1 as a result

E1

or $\frac{1}{12} = \frac{1}{k-500}$

k = 512 as a result

1

(b) Mean, $\mu = \frac{512+500}{2} = 506$

CAO

B1

Variance, $\sigma^2 = \frac{(512-500)^2}{12} = 12$

Use of; allow even if called standard deviation

M1

Thus standard deviation, $\sigma = \sqrt{12} = 3.46$

CAO/AWRT

A1

3

(c) (i) $P(X < 509) = \frac{509-500}{12}$

Attempt at area of rectangle below 509 or of integral from 500 to 509

M1

$\frac{3}{4} = 0.75$

CAO

M1

2

(ii) $P(X > \mu - \sigma) = \frac{512-(\mu-\sigma)}{12}$

Attempt at area of rectangle above $(\mu - \sigma)$ or of integral from $(\mu - \sigma)$ to 512 allow use of value of σ^2 only if subsequent answer is stated as 1

M1

$\frac{512-(506-\sqrt{12})}{12}$

√ on μ and σ in formula or integral; not on σ^2

A1f.t.

$\frac{6+\sqrt{12}}{12} = 0.788$ to 0.789

AWFW

A1

3

[9]

Q17.

(a) *c*/2

c/2 cao

B1

1

(b) $E(X^2) = \int_{-c}^{2c} \frac{1}{3c} x^2 \,dx = \left[\frac{1}{3c}\frac{x^3}{3}\right]_{-c}^{2c}$

Any correct expression – ignore limits

M1

Any correct integration

M1

Correct method apart from numerical/algebraic slips

m1

$= \frac{1}{9c}[8c^3 - c^3] = c^2$

*Completely correct method **ag***

A1

4

(c) Variance = $c^2 - (c/2)^2 = 3c^2/4$

Correct method their answer to (a) – allow variance if called variance

M1

standard deviation = $c\sqrt{3/4} = 0.866c$

Allow any correct method – allow variance if called variance

m1

$c\sqrt{3/4}$ acf or 0.866c (0.866c ~ 0.867c)

A1

3

(d) (i) 22 is estimate of *c*/2.

Method for c – their answer to (a)

M1

Estimated value of *c* is 44

44 cao – may be implied later

A1

Estimated standard deviation of *X* is

Method for s.d. their answer to (c)

m1

$44 \times \sqrt{3/4} = 38.1$

38.1 (38 ~ 38.2) allow 22√ 3

A1

4

(ii) Minimum weight is 2000 – *c* grams

Method their c

M1

estimated by 1956 grams.

1956grams or 1.956 kg – allow1960 or
1.96 units required

A1

2

[14]

Q18.

(a) $\int_{-6}^{14} k\,dy = [ky]_{-6}^{14}$ $= k(14 - (-6)) = 20k$

Allow by diagram – generous

M1

$20k = 1$ $k = 0.05$

0.05 ***ag***

A1

2

(b) 4

4 cao

B1

1

(c) (i) $\int_{-6}^{14} 0.05y^2\,dy = \left[\frac{0.05y^3}{3}\right]_{-6}^{14}$

Any correct expression for E(Y²) – ignore limits
Correct method – arithmetic/sign errors only

M1 m1

$= 45.733 - (-3.6) = 49.3$

49.333 (49.3 – 49.35)

A1

3

(ii) Var(Y) = 49.3333 – 4^2 = 33.3333

Method for Var – their E(Y^2)

M1

s.d. = $\sqrt{33.3333}$ = 5.77

Requires all previous method marks in (c)

5.77(5.77 – 5.775)

m1 A1

3

(d) (i) 0.3

0.3 acf

B1

1

(ii) $\int_{-2}^{2} 0.05 dy = [0.05y]_{-2}^{2}$

Allow diagram

M1

= 0.2

0.2 ***ag***

A1

2

(iii) 0.6

Allow diagram

0.6 acf

M1 A1

2

[14]

Q19.

(a) (i) c/2

cao

B1

(ii) $\int_0^c x^2 / c \, dx = [x^3 / 3c]_0^c = c^2 / 3$

any correct expression - allow no limits

correct evaluation of a definite integral

M1 M1

variance = $c^2/3 - (c/2)^2 = c^2/12$

for variance - their E(X) and E(X^2)

requires all previous M marks
(answer given)

M1 A1
5

(b) $\int_{c/4}^{c} \frac{1}{x} dx = \left[\frac{x}{c}\right]_{c/4}^{c}$

method - integration or diagram
¾ cao acf

M1 A1
2

(c) (i) 14 is estimate of *c*/2
any correct expression involving estimate of c

M1

28 is estimate of *c*

A1

28/√12 = 8.08 is estimate of standard

m1

deviation of X
for standard deviation
8.08 ~ 8.1

A1

(ii) 1028 grams
or 1030

B1ft
5

[12]

Examiner reports

Q2.

A reasonable proportion of students appreciated that a full explanation of why a = 27 required some mention of zero waiting time. Many wasted time with prolonged calculation of b, as they did in part (b), where formulae from the booklet could have been used. For those students who did use the variance formula, it was surprising how many wrote down the correct formula, but then in calculating the answer the square sign was lost or the $\frac{1}{12}$ became a ½ . Whilst many students found part (c) straightforward, in addition to the expected incorrect use of a 10-minute interval, a one-sided interval of five or sometimes even three minutes, was frequently seen.

Q3.

On the whole, except for the more able candidates, the attempts at this question were very disappointing, especially as a similar question had appeared on a past paper. Using formulae given in the formula booklet and finding the area of a rectangle is all that was required, yet it seemed to be beyond the vast majority of candidates. In part (a), there seemed to be a lack of understanding of the concept of rounding errors. Although many candidates gave the correct lower value as 21.05, there were far too many who incorrectly gave the upper value as 21.14. Thus the correct values of 21.05 and 21.15 were rarely seen. A sketch indicating 21.1 ± 0.05 may have been useful. In part (b), the vast majority of candidates tried to work with widths and failed altogether to consider the distribution of rounding errors, X. Consequently, the overwhelming majority incorrectly thought that $\mathrm{E}(X)$ = $\frac{1}{2}$ (21.05 + 21.15) = 21.1 and that Var(X) = $\frac{1}{12}$ (21.15 – 21.05)^2. Although the latter often led to the correct numerical exact answer for the standard deviation, this method gained no credit. Very few diagrams of the required rectangular distribution of errors were seen. Some candidates, having obtained the correct variance by a valid method, failed either to go on to state the value of the standard deviation or did not comply with the request for an exact answer, often giving 0.0289. In part (c), although most candidates worked with an interval of 0.04, they were then unable to find the correct value of p = 0.4 with 0.04 × $\frac{1}{10}$ = 0.004 often seen.

Q4.

In part (a)(i), it was expected that candidates would use ‘area of rectangle = 1’. This was not always the case, with integration methods again being employed where they are totally unnecessary and time consuming. Those who did manage to write down area = 10u × 0π011 = sadly then often seemed unable to complete the required algebraic manipulation in order to obtain the given answer.

In part (a)(ii), most candidates managed to correctly find E(X) and Var (X) by using the suggested formulae. Some however did not leave their answers in terms of as requested and consequently lost marks.

Part (a)(iii) was very poorly answered except by the most able candidates. Far too many candidates either did not know the formula $C = \pi D$, or failed to realise that this formula needed to be used, and so simply attempted to find $E\left(X+\frac{10}{X}\right)$ and $\mathrm{Var}\left(X+\frac{10}{X}\right)$, often incorrectly. However, part (b) was usually very well done.

The vast majority of candidates realised that, as the variance of the given distribution was known, a z-value (not a t-value) had to be used.

Q5.

Too many candidates seemed to think that a request for a 'sketch' entitled them to draw a very poor quality graph. A horizontal straight line was expected but not always seen. Some candidates failed to draw a graph of the given function, instead using the given value of $k = \frac{1}{2}$ to draw a graph of f(x) = $\begin{cases} \frac{1}{2} & -\frac{3}{2} \le x \le \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$, which obviously lost credit. In part(a)(ii), where the answer was given, too little method was frequently seen for the candidate to be awarded full marks.

In part (b), the majority of candidates used the formula E(X) = $\frac{1}{2}$ (a + b) to correctly calculate the mean. However, the calculation of the **exact** numerical value of the standard deviation was less well done. Many candidates found the variance to be $\frac{1}{3}$ but then either failed to state the value of the standard deviation or gave a decimal value (usually 0.577) where an **exact** value (equivalent to $\frac{1}{\sqrt{3}}$) was required.

In part (c)(i), many found the correct answer of $\frac{3}{8}$. Unfortunately there were several candidates who used this value in working out their answer to part (c)(ii). Consequently, an answer of $\frac{5}{8}$ was a popular incorrect answer, along with the inevitable zero! However, most candidates did realise that $\mathrm{P}\left(X \ne -\frac{1}{4}\right) = 1 - \mathrm{P}\left(X = -\frac{1}{4}\right)$ = 1 – 0 = 1.

Q6.

Part (a) was usually very well done, with many correct answers of $k = \frac{1}{a+b}$ seen.

Unfortunately, there were some candidates who simply quoted the formula $k = \frac{1}{b-a}$ from page 11 of the formulae booklet without realising that this was based on the interval [a, b], whereas this question indicated an interval [a, −b]. Integration methods were also seen, which usually resulted in the correct expression for k. It was pleasing to see that the vast majority of candidates realised that $\mathrm{E}(t) = \int_{-a}^{b} kt\,\mathrm{d}t$, with most then integrating correctly.

Although the correct substitution of the given limits leading to $\frac{1}{2}k(b^2 - a^2)$ was often seen, many candidates then simply quoted the given answer without any further working. This resulted in a loss of marks. It was expected that candidates would factorise to get $b^2 - a^2 = (b - a)(b + a)$ and perform a cancellation before stating the final given result. When an answer is given in the question, it is essential for candidates to take all possible care to show all relevant working.

In part (b)(i), although the correct answer E(T) = 1 was seen, many candidates thought that E(T) = 5. This wrong answer usually resulted from candidates not realising that their answer to part (a)(ii), namely E(T) = $\frac{1}{2}(b-a)$, was based on the given interval $[-a, b]$ and so $[-4, 6]$ meant that $a = 4$ and **not**, as they assumed, $a = -4$. Although the application of integration was also used here to obtain, in most cases, the correct result, it was not the most efficient method and should not be encouraged when dealing with rectangular distributions.

In part (b)(ii), those candidates who drew a diagram were then, more often than not, able to write down the correct answer of 0.4. However, although many candidates were able to calculate P(T < –3) = 0.1 and P(T > 3) = 0.3, they then seemed to be at a loss as to what to do next. They failed to realise that the word ‘or’ was asking them to add these two probabilities together. Some candidates calculated P(T < –3) = 0.1 or P(T > 3) = 0.3, obviously believing that they had a choice, whilst others incorrectly thought that P(T > 3) = P($T \geq 4$) = 0.2. Candidates who drew a diagram usually then managed to write down the correct answers to all of part (b) with very little or no further working.

Q7.

Part (a) caused the most problems in this question. Although many candidates were able to demonstrate why f(x) =10 for $-0.5 \leq x \leq 0.5$, they were unable to indicate why the values of X had to lie in the given range. Part (b) was often done correctly, if not always by the most efficient method. For this rectangular distribution, there were far too many candidates still using integration, rather than a sketch diagram and simple geometry, to reach the required result. It was expected that candidates would use the formulae from page 11 of the booklet provided to answer part (c). Although many did, there were those who attempted integration methods with little success. Some candidates who managed to find the variance then did not go on to find the standard deviation.

Q8.

Here again a wide range of marks was seen. All but the weakest candidates found part (a) accessible and, by substituting a and $a + k$ into formulae from the booklet, were able to score four marks. Whilst many candidates scored the two marks in part (b), a significant proportion carelessly equated [(–6 + 24) – (–6)] to 30.

Q9.

More able candidates scored 10 marks on this question but others only scored about half this number.

In part (a) knowledge of the formulae for mean and variance was good. However about half of the candidates then showed that a = 12 and b = 30 satisfied the requirements; this was not considered a proof. More able candidates set up and then solved two simple simultaneous equations, but almost all failed to realise that $\sqrt{324} = \pm 18$ and lost one of the five marks available.

Parts (b) and (c) proved to be very accessible to many candidates. However, some failed to recognise, in part (b), that the lower limit was 12 so that the height was $\frac{1}{18}$, not $\frac{1}{30}$. This error was carried forward into part (c), although P(X < 16.5) was often deduced from substitution in the given algebraic expression. A minority of candidates, particularly in part (b), attempted to use a normal distribution, for which no marks were awarded.

Q10.

Many candidates scored most of the first five marks. Where marks were lost it was usually for finding P(X > 200) or using an incorrect expression for the variance despite it being listed in the booklet supplied.

Answers to part (b) scored in the main only one mark for 'n large' or 'Central Limit Theorem' so 'normal'. Full marks also required 'mean = 185' and 'variance = 9'.

Q11.

It was expected that most candidates would find part (a)(i) very accessible. As stated earlier in the report, this was not the case due to poor algebraic skills. As a result, many candidates scored at most one of the two marks in part (a)(ii) for a correct method, not for a fiddled answer. Again, in part (b), many candidates lost the mark because they were unable to solve $19c^2 = 171$ correctly. In part (d)(i), even with follow-through marks, candidates simply could not find the 'correct' probability; some even tried using a normal distribution. In part (d)(ii), many candidates simply evaluated (7c – c) × 0.25, apparently forgetting that the distribution started at c. Having said all this, some candidates scored all eleven marks through a good knowledge of the relevant statistics and the ability to handle relatively simple algebra.

Q12.

It was disappointing to see the large number of incorrect answers to part (a); this despite the relevant formulae being listed on the supplied booklet. Additionally, candidates often simplified $\frac{(4c)^2}{12}$ to $\frac{4c^2}{12}$ or $\frac{16c}{12}$.

In general, only the best candidates made any headway in part (b), but even some of these were hampered in using Var (X) = E(X^2) – (E(X))^2 by their algebraic errors in part (a). A small number of candidates equated $\frac{28}{3}$ to $\int_{-c}^{3c} \frac{x^2}{4c} \, dx$,often with complete success.

Answers to part (c)(i) were almost always correct and it was pleasing to see good attempts at calculating the probability in part (c)(ii). However, although most associated diagrams showed a lower limit of –2, many candidates apparently ignored the modulus signs and calculated P(0 < X < 2.5).

Q13.

This was considered to be a fairly straightforward question on the rectangular distribution but candidates' disappointing responses suggested otherwise. Many candidates only scored marks in part (b), whilst a considerable number omitted the entire question. It remains concerning that so many candidates apparently find this topic so difficult.

In part (a), those candidates who used the supplied booklet to find the formulae for the mean of a rectangular distribution and then match a and b to the given form before equating it to 9 were usually successful. Reverse arguments were rarely sufficiently complete to score full marks as were most attempts at integration. Using b = 6 to show that the total area was then unity scored no marks, as did attempts at a discrete distribution. Answers to part (b) were much better and often correct even following no or incomplete attempts at part (a). A minority of candidates either found P(T > 15) or ignored the area for P(T < 0).

Q14.

In part (a), a significant number of candidates used a and k rather than a and $a + k$, or did not simplify their correct expressions. Those candidates who obtained correct simplified expressions in part (a) generally scored the three marks in part (b). Here about half the solutions involved solving 2a + k = 42 and k^2= 36 for k and a, while the others showed that a = 18 and k = 6 satisfied E(X) = 21 and Var(X) = 3. Correct answers to part (c) were the norm, with most candidates recognising that the answer equated to the rectangular area of (24 – 20) x $\frac{1}{6}$.

Q15.

Most candidates scored the 2 marks in part (a), though a few chose to divide by 2 rather than 12 when finding the variance. It was pleasing to see the large number of correct answers to parts (b) and (c)(i); the usual incorrect values were $\frac{1}{6}$ or $\frac{1}{7}$. In answering part (c)(ii), candidates tended to use their value of σ^2, rather than σ and/or failed to find an actual area by not using their answer to part (b). Those candidates who sketched a diagram were often the more successful.

Q16.

Attempts at this question on a rectangular distribution were much improved over answers on previous papers. In part (a), almost all candidates reasoned correctly as to why k = 12 and then scored the 3 marks in part (b), though a minority used $\sigma^2 = \frac{1}{12}(b - a) = 1$ or σ = 12.

Answers to part (c) (i) were sound and, in the main, scored full marks. However, in part (c) (ii), many candidates used $\frac{(\mu - \sigma) - 500}{12}$, often obtaining the correct value of 0.211, but then did not find the required answer of 1 – 0.211 = 0.789.

Q17.

This question caused problems for many candidates.

In part (a) candidates commonly engaged in lengthy attempts to show the result, rather than simply writing it down.

Integration, which defeated many candidates, was only required in part (b) but candidates who could not make a reasonable attempt at part (b) rarely continued with the rest of the question.

Most candidates were well prepared for part (a) and answered it competently. Parts (b) and (c) caused more problems. In part (b) some candidates were defeated by the problem of comparing a positive test statistic with a negative critical value.

Part (c) caused even more problems and was omitted by some candidates. Not all of those who did attempt it realised that a null hypothesis must contain an equals sign.

Q18.

Most candidates used integration throughout this question. Apart from part (c)(i), methods based on diagrams or quoting formulae were equally acceptable. Part (d) caused the most problems, with some apparently unsure about the meaning of 'magnitude'.

Q19.

Some found this question easy but many were put off by the integration. Integration was only needed for part (a)(ii). Part (a)(i) required no proof; part (b) could be answered using a diagram and part (c) by using the earlier results. However, those who were unable to deal with the integration were generally put off from attempting the rest of the question.